STATS 402 - Interdisciplinary Data Analysis

Short-Horizon Temporal Optical Flow with Physics-Informed Consistency

Milestone Report: Stage 3

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1. Current Status of the Project

1.1 Pipeline and Architecture Implementation

At the current stage, we have completed the full deep learning pipeline for short-horizon temporal optical flow estimation with physics-informed consistency. Building on the implementations in Stage 2, the current framework now includes:

- Pairwise Flow Encoder
- Joint Visual-Motion Fusion Encoder
- U-Shape Fourier Neural Operator (UNO) Model with Auxiliary Input-Output Modules
- PWC-Style Decoder
- A Self-Supervised training framework
- Training Pipeline for both local device and DKU Computing Cluster (DKUCC)

Figure 1 and 2 show the current full pipeline, which now includes all components that the proposal specified. It is notable that as discussed in the Milestone Stage 2 report, we conduct experiments on UNO implementation on different stages in our full pipeline. Therefore, we now offer three variants, which are

a. Original formulation: standalone vanilla UNO after the Joint Visual-Motion Fusion Encoder, before the PWC-style decoder;
b. Early integration of the module after the Pairwise Flow Encoder, before the Joint Visual-Motion Fusion Encoder; and
c. Late integration after the Joint Visual-Motion Fusion Encoder and before the PWC-style decoder.

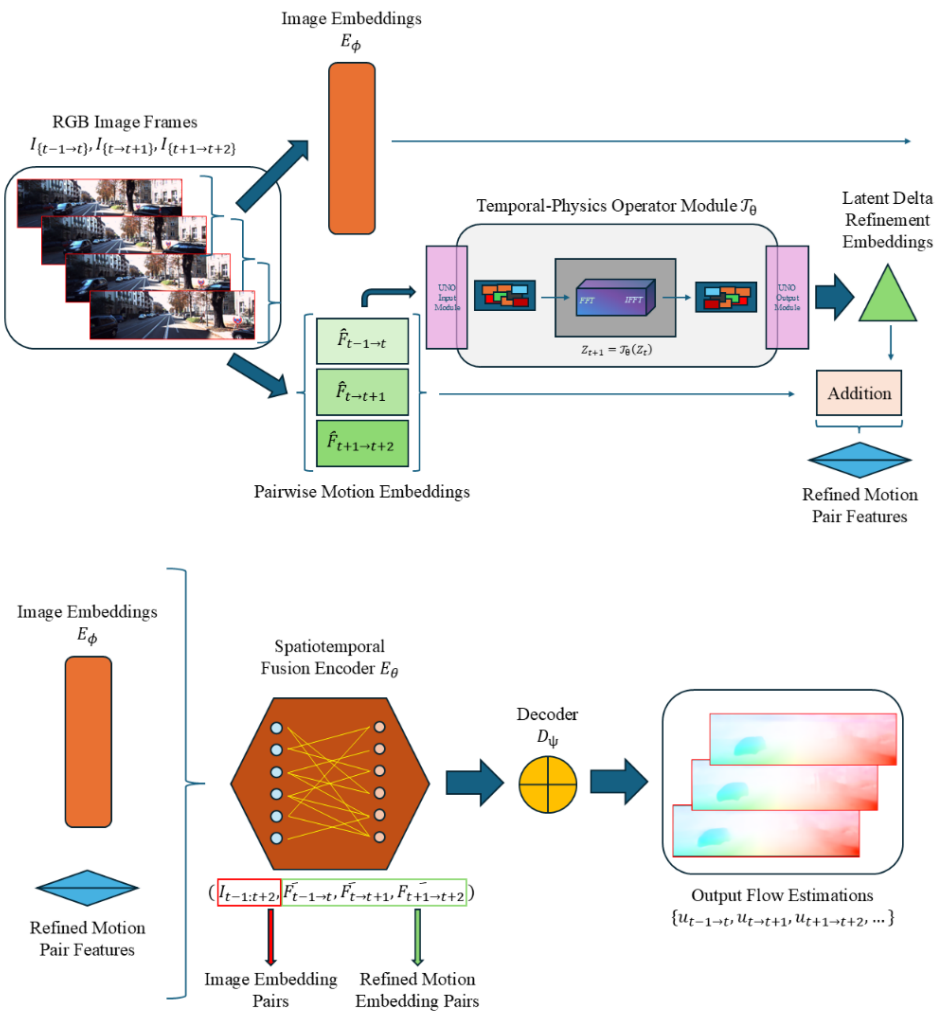

Figure 1: Current full pipeline with variant b. Early integration of UNO after the Pairwise Flow Encoder, before the Joint Visual-Motion Fusion Encoder

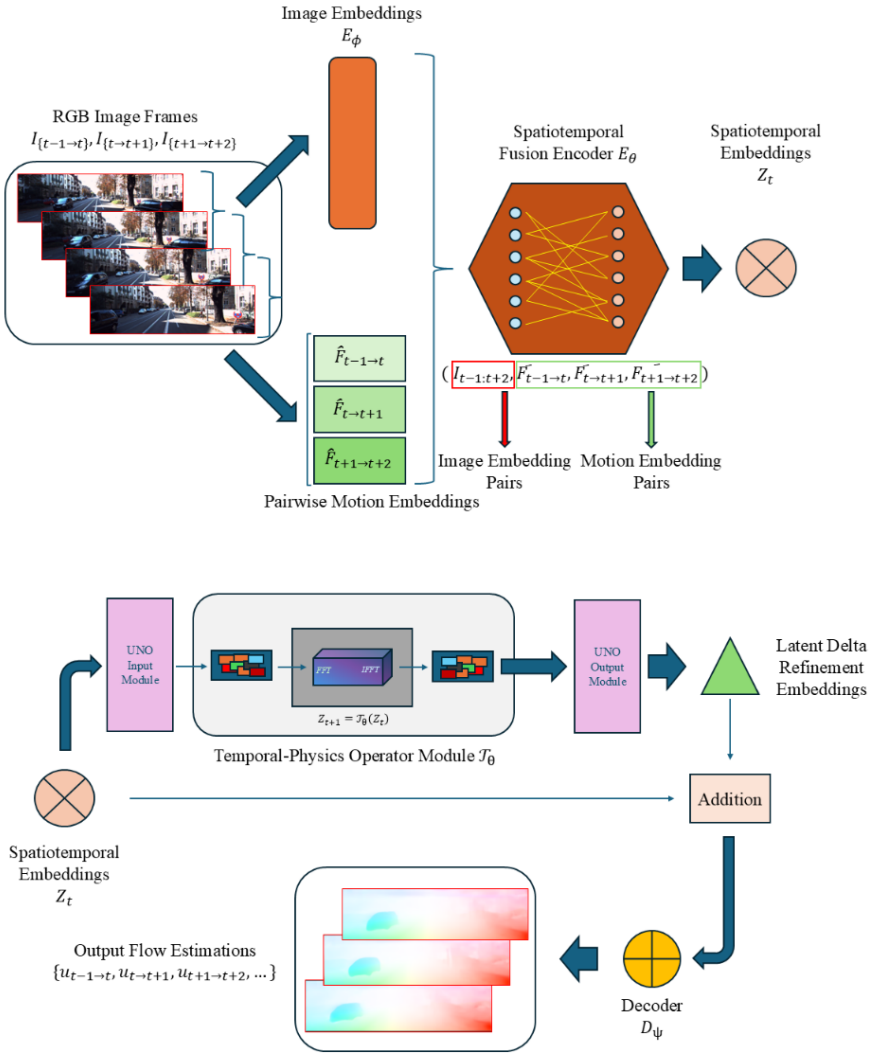

Figure 2: Current full pipeline with variant c. Late integration after the Joint Visual-Motion Fusion Encoder and before the PWC-style Decoder.

We will elaborate on technical details, performance, and systematic suggestions on both variants in the later section on Technical Adjustments.

1.2 Technical Adjustments

1.2.1 Structural Adjustments

As proposed in the Proposal, we planned to incorporate a UNO module in the later stages of encoding. The module should take in a combined down-sampled spatiotemporal embedding. It should extract meaningful image-motion features, and outputs another latent embedding that serves as the input of the decoder module. The original formulation is shown in Figure 3.

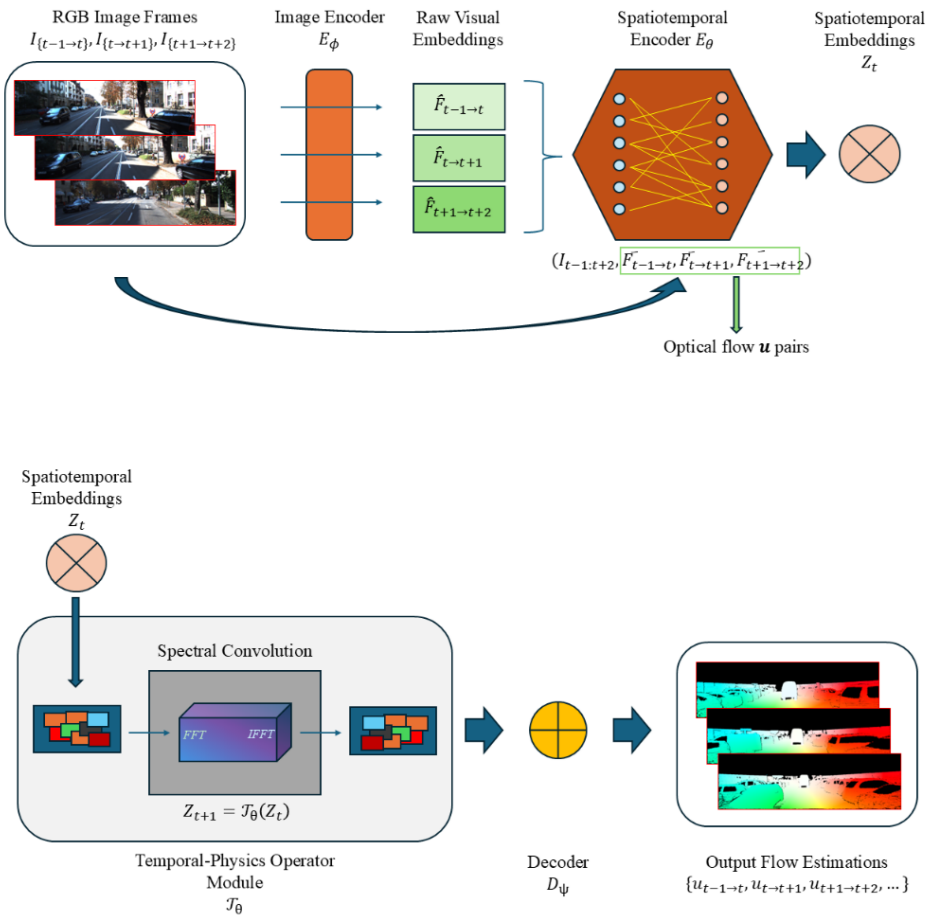

Figure 3: Original formulation of our framework (variant a). The UNO module takes embeddings from the previous Fusion module, and outputs to the Decoder

We conducted experiments by code implementation and training. We first tried the original formulation, which simply takes the fused image-motion embeddings from the previous encoders, and directly output to the decoder. Even though improvements in temporal consistency improved somewhat, it showed no significant progress.

We suspect the previous lack of improvement is caused by the down sampled feature layers. During processing, each input image or ground truth flows is down sampled $4\times$ via convolution. After this down sampling, the bottleneck data size is 44×152, compared to its original 352×1216. This has significantly decreased the amount of information passed to the operator, where it fails at retrieving sufficient, physically meaning information for pattern learning. To cope with the lack of information, we shifted to modifying the Neural Operator module to better align with the framework

behavior. This introduces our modifications on UNO integration and functionality. Inspired by ResNet, we instead integrate the operator module by making this a residual refinement module that takes information before the down sampling concludes, and allows it to learn flow behaviors at a larger input data scale (He et al., 2016).

As mentioned in Milestone 2, we also encounter estimation anomaly on the sky regions. Similar to the bottleneck situation in our original formulation with the neural operator, this can also be attributed to the result of down-scaled convolution. To alleviate this problem, we borrow the idea of skip connection in UNet (Ronneberger et al., 2015). Here, we keep latent features at certain down sampling stages (since we only down sampled by 4 ×, there is no need to keep all of them), and feed the up sampled estimations with these features on the same level. Nevertheless, to keep a baseline performance, we kept this functionality optional.

1.2.2 Adjustments on Loss Functions

On this stage, we further refine the loss function to address several limitations observed in Stage 2, including weak supervision in low-texture regions, accumulation of motion errors across frames, and insufficient geometric consistency. The current overall loss function is defined as follows:

$$\mathcal{L} = \mathcal{L}_{flow} + \lambda_{photo}\mathcal{L}_{photo} + \lambda_{acc}\mathcal{L}_{acc_photo} + \lambda_{fb}\mathcal{L}_{fb} + \lambda_{temp}\mathcal{L}_{temp} + \lambda_{smooth}\mathcal{L}_{smooth}$$

Except for the temporal loss, we made modifications to every other parts in the prospect of improving model performance. We elaborate on the specifications of each loss function below:

(1) Supervised Flow Loss:

$$\mathcal{L}_{flow} = \frac{\sum \|F - F^{gt}\| \cdot \left(1 + \lambda_{edge} \cdot edge\right)}{\sum\left(1 + \lambda_{edge} \cdot edge\right)}$$

Following Stage 2, we now introduce an edge-aware weighting mechanism into the standard endpoint error (EPE) loss. The motivation of this modification is to improve motion estimation near object boundaries. In practical scenarios, motion discontinuities often coincide with strong image

gradients. Therefore, by assigning larger weights to pixels with higher gradient magnitude, the model is now encouraged to focus more on regions where accurate motion estimation is more challenging.

(2) Multi-frame Photometric Loss:

$$\mathcal{L}_{photo} = \alpha L_1 + \beta SSIM + \gamma Census$$

The photometric loss follows the self-supervised reconstruction framework introduced in Stage 2. Instead of relying on a single metric, we combineL_1, SSIM, and Census terms to improve robustness against illumination changes and local texture variations. This formulation is widely adopted in optical flow methods such as UnFlow (Meister et al., 2017), UFlow(Sun et al., 2026) (Jonschkowski et al., 2020), and U2Flow (Sun et al., 2026).

(3) Accumulated Photometric Loss (Possible cite: 19 SelfFlow)

$$F_{t\to t+2}(x) = F_{t\to t+1}(x) + F_{t+1\to t+2}\big(x + F_{t\to t+1}(x)\big)$$

$$\mathcal{L}_{acc_flow} = \frac{\sum \mathcal{L}_{acc}(t, t+2) \cdot W_{t,t+2}}{\sum W_{t,t+2}}$$

To further improve temporal consistency, we introduce an accumulated photometric loss defined over two-step frame transitions. Specifically, the flow from frame $t$to $t+2$is obtained by composing two consecutive flows. The reconstructed image is then compared using the same photometric loss formulation as defined previously.

(4) Forward–Backward Consistency Loss

$$\mathcal{L}_{fb} = \sum \| F_{fw}(x) + F_{bw}\left(x + F_{fw}(x)\right)$$

This term enforces geometric consistency between forward and backward flows. Ideally, a pixel mapped forward and then backward should return to its original position. This constraint is commonly used in self-supervised optical flow methods such as UnFlow and UFlow.

(5) Edge-awareness Smoothness Loss

$$\mathcal{L}_{smooth} = \frac{1}{T-1}\sum_{t=1}^{T-1}\sum\left(|\nabla F_t| \cdot e^{-|\nabla I_t|}\right)$$

Compared to Stage 2, where the smoothness loss is applied to a single flow prediction, we extend this formulation to all predicted flows within the sequence. Specifically, the loss is computed for each adjacent flow F_t, and the final value is obtained by averaging across all $T - 1$flow predictions. In our setting, this corresponds to averaging over three flows. This modification ensures consistent spatial regularization across time while maintaining a comparable loss scale.

This formulation follows commonly used edge-aware smoothness constraints in optical flow and related tasks, such as UnFlow (Meister et al., 2017) and Monodepth (Godard et al., 2018), where gradient-based weighting is used to preserve motion boundaries.

1.2.3 Downstream Ablation

As indicated in our Milestone 2 Report, we will introduce a bounding box based downstream task. This task is intended to evaluate if the machinery vision is capable of recognizing and identifying our flow estimations. Fortunately, KITTI dataset provides their official object labeling, which is aligned from RGB images with their flow ground truth. Thus, we are able to utilize them to develop a lightweight recognition module.

The formulation of the framework is relatively simple. We utilize pre-defined backbones retrieved from the torchvision library, and truncate its first-layer input shape to match our flow estimation shapes, which completes the simple detector. We offer two setups for the backbone: a lighter MobileNet, and a slightly heavier but more capacitated ResNet50. Since the optical flows consists of two channels, the input size to the detector adds a one-channel bounding-box layer, which results in a total 3-channel shape.

We conducted training on the ResNet50 backbone, and conducted two experiments varying input flow types:

(1) Ground Truth Optical Flow as Input

This version retrieves ground truth optical flows from the training batches as the input to the detector. However, as a downstream inference task, estimated optical flows would be fed as input, which creates an out-of-

distribution condition, leading to certain underperformance under the setup.

(2) Estimated Optical Flow as Input

This version intends to mitigate the out-of-distribution problem caused by the mismatch of data between training and inference. However, to be rigorous and honest in evaluating model performances, this version is deprecated.

2. Demonstration of Current Results

We demonstrate results on all variants of our model, which includes vanilla UNO implementation, early UNO incorporation, and late UNO incorporation. All results are based on the same hyperparameters, model structure and other configurations, except for varying UNO implementation strategies.

We first present the quantitative evaluation of all model variants on the training set.

We follow the standard evaluation protocol used in optical flow benchmarks such as KITTI and RAFT. Even though we utilize a series of loss functions to regularize the model, three most characteristic and diagnostic metrics in this report: End-Point Error (EPE), per-image EPE (Fl-EPE), and outlier ratio (Fl-all%). Specifically:

- EPE measures the average magnitude of the difference between predicted flow and ground truth. Lower values indicate better accuracy.
- Fl-EPE computes the average EPE per image and then averages across all samples. This reduces bias from image size and provides a more stable estimate of performance. Lower values indicate better accuracy.
- Fl-all% measures the percentage of pixels whose error exceeds a predefined threshold. A pixel is considered incorrect if its error is larger than 3 pixels and exceeds 5 percent relative error. This metric evaluates robustness by focusing on large errors. Lower values indicate better performance.

The Early Integration model achieves the lowest EPE and Fl-epe, indicating the best average accuracy. This may be because the operator is applied earlier and directly influences feature extraction.

The Original (Standalone Operator) achieves the lowest Fl-all, meaning it produces fewer large errors. A possible reason is that it makes more conservative predictions without relying on fused features.

The Late Integration (Refined) model shows intermediate performance. Since the operator is applied after fusion as a refinement step, it benefits from structured features while still adjusting the final output.

The specific values are shown in Table 1.

Model Variant	EPE ↓	F1-EPE ↓	F1-all% ↓
Standalone	2.9234	2.7889	**20.59**
Early Integration		**2.5596**	22.11
Late Integration	2.7647	2.6673	23.30

Table 1. Specific values for EPE, F1-EPE and F1-all% for all three UNO integration variants

We also include the signature training dynamics of the three framework variants. As shown in Figure 4, the three plots show Weighted EPE Loss, Latent Space Change, and Temporal Consistency Loss from left to right.

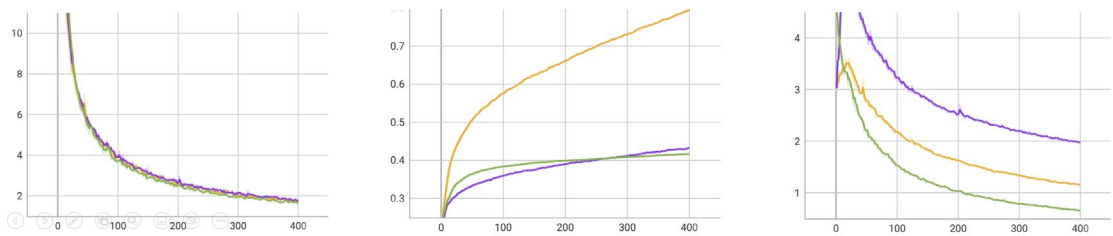

Figure 4. Training dynamics of the 3 operator integration variants from tensorboard. Yellow refers to late integration; purple refers to early integration; and green refers to standalone

- From the **Weighted EPE Loss** plot, all three models show similar decreasing trend and converge to close values. This suggests that all variants can learn reasonable supervised flow estimation.
- From the **Latent Space Change** plot, the Late Integration model shows much larger latent updates than the other two variants. This indicates that the Late Integration variant has more impact on the main pipeline of our model. In this version, the operator is applied after feature fusion as a refinement module, thus it directly adjusts the fused representation. The Early Integration and

Original Standalone Operator show similarly smaller values in this aspect, suggesting more conservative latent updates.

- From the **Temporal Consistency Loss** plot, the Original Standalone Operator achieves the lowest temporal loss, followed by Late Integration, while Early Integration remains the highest. This shows that our original formulation achieves the best at preserving temporal smoothness, underlying the ability of UNO modules to effectively learn spatiotemporal evolution of objects of interest. A possible reason for the worsened performance on the other two variants may be that early features are less structured, so applying the operator too early may introduce temporal fluctuation. In contrast, the standalone and late-integration designs operate on more processed representations, which may help produce smoother short-horizon motion behavior.

2.1 Variant a: Standalone Vanilla UNO Integration (Original Plan)

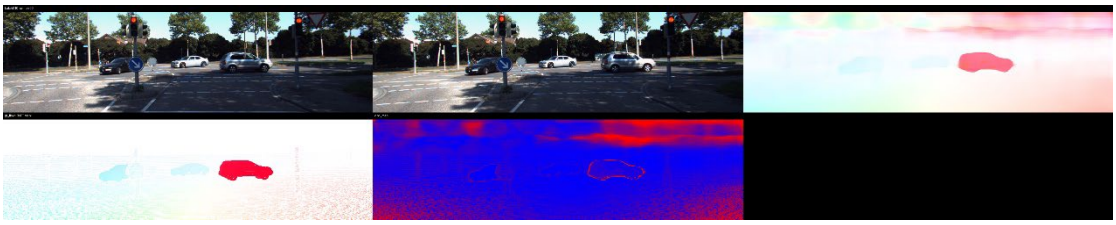

Figure5: Visualization of optical flow predictions for the Standalone integration

Figure 6: Visualization of optical flow predictions for the Standalone integration on self-supervised short-horizon flow

Text.

2.2 Variant b: Early UNO Integration

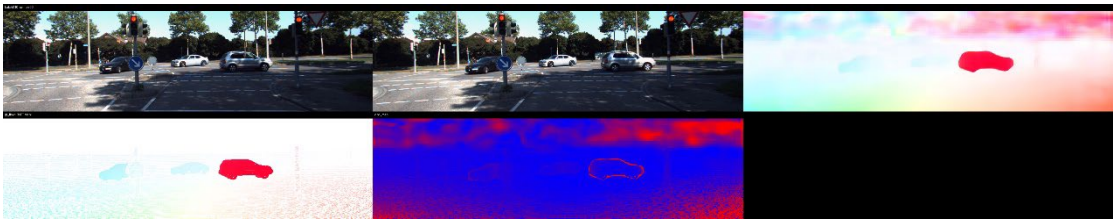

Figure7: Visualization of optical flow predictions for the Early integration

Figure 8: Visualization of optical flow predictions for the Early integration on self-supervised short-horizon flow

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2.3 Variant c: Late UNO Integration

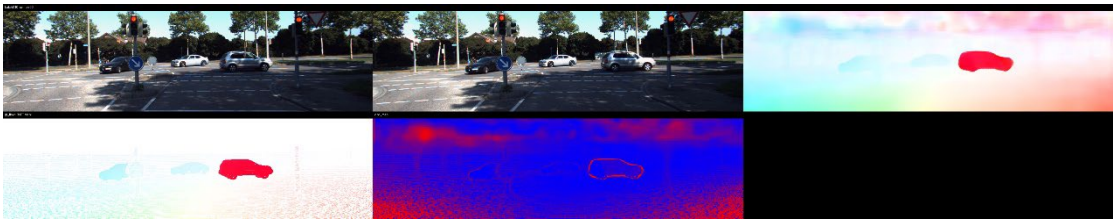

Figure 9: Visualization of optical flow predictions for the Late integration

Figure 10: Visualization of optical flow predictions for the Late integration on self-supervised short-horizon flow

Text.

2.4 Downstream Bounding Box Detection Task

Two experiments are conducted with only differing model training input. Both experiments are trained for 50 epochs, where the model has fully converged. The score threshold for visualization is 0.5 (out of 1).

(1) Ground Truth Optical Flow as Detector Input (Figure x and y).

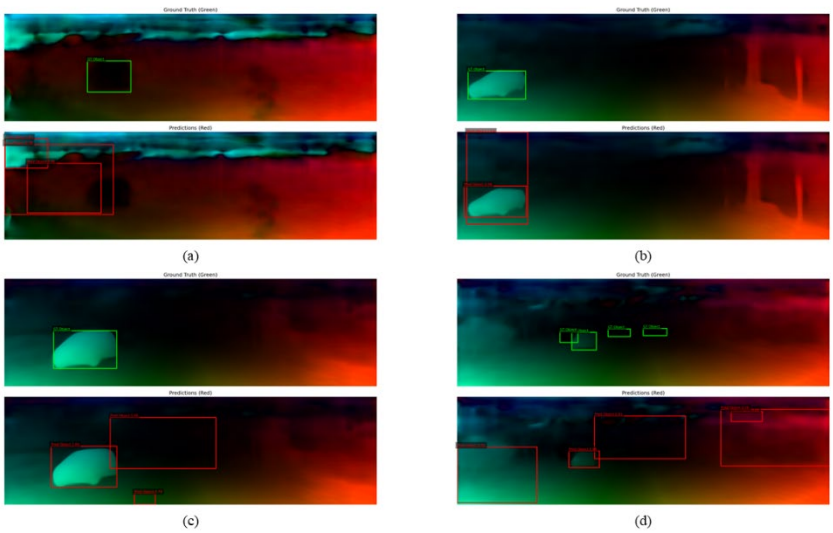

Figure x. Detection model inference on model trained with **ground truth** optical flow as detector input.

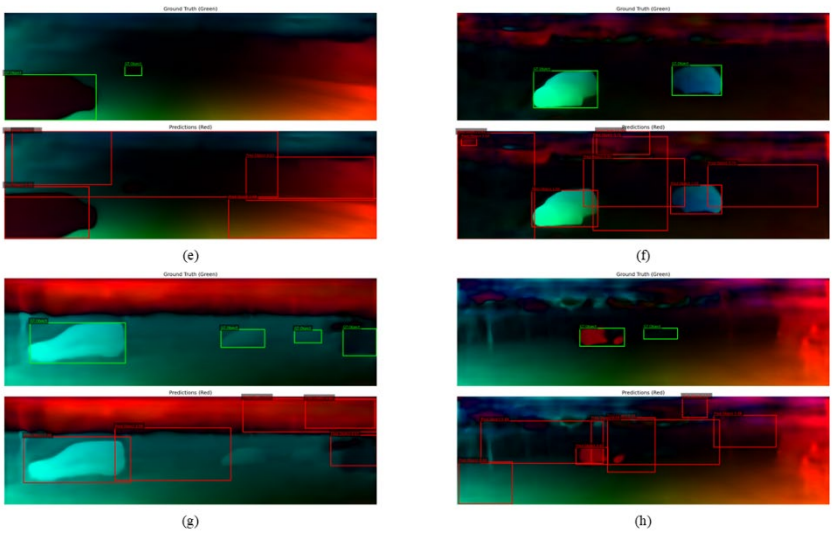

Figure y. Detection model inference on model trained with **ground truth** optical flow as detector input. More samples.

As expected, the out-of-distribution challenge was encountered during training and inference. As illustrated in the resulting visualizations, the model can, in general, recognize the objects of concern with a relatively high score. However, they tend to produce false positives, where excessive incorrect boxes are bounded, often recognizing flows produced by background ego motion. Nevertheless, as the detector is only intended to be exposed to estimated flows, this distribution shift can be easily mitigated by stacking the ground-truth bounding box mask with estimated flow estimations as model input, which introduces our next experiment.

(2) Estimated Optical Flow as Input (Figure x and y)

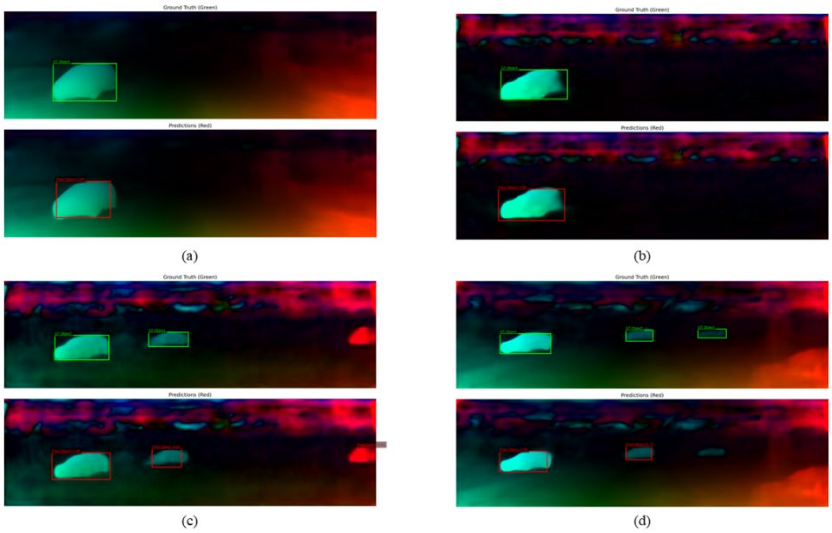

Figure x. Detection model inference on model trained with **estimated** optical flow as detector input.

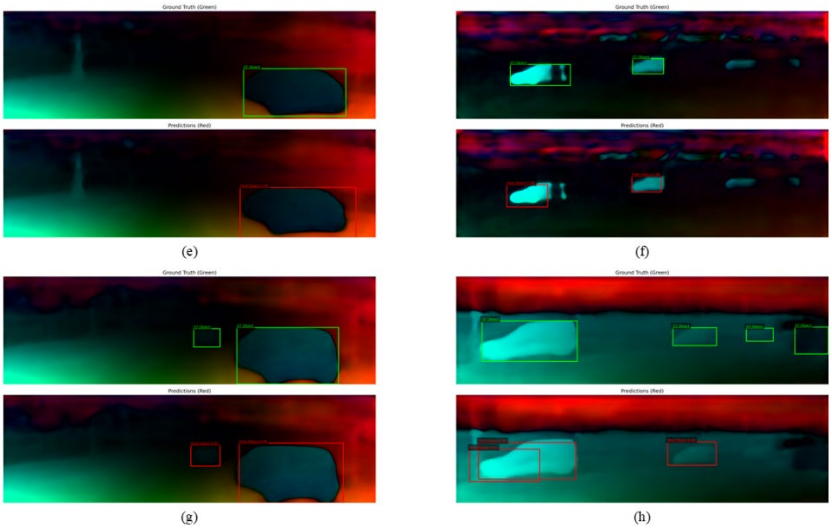

Figure y. Detection model inference on model trained with **estimated** optical flow as detector input. More samples.

Training conducted using estimated optical flow as decoder input improved inference results. As shown in the figures, most regions of interest are identifiable by the detector module, while a relatively small proportion of samples fail to fully detect the object. In this variant, the model detected no false positives, whereas insufficient detection could be attributed to the performance of our estimation framework.

Generally, both experiment show that machinery image successfully recognize the majority of flow motions estimated by our framework. This shows that our flow estimation model not only achieves an acceptable performance on producing flows, but the results are also recognizable by deep learning-based methods.

3. Future Plans

On this stage, we have completed the plans listed in Milestone 1 Proposal and Milestone 2 Report. Nevertheless, future plans are yet present in terms of final submission, and long-term development prospect.

3.1 Final Submission

The current working directory needs cleaning up and better formatting for final submission of our full code. This step include structuring documents and files into a more professional manner, as well as polishing README.md to better articulate our project intention, objective, and results.

It is also worth noting that current results have not yet been conducted on test sets offered by the KITTI dataset. For final submission and evaluation, we will hold validation runs on the official test set, where no ground truth of label is visible to developers. This would increase the robustness of our qualitative evaluation later on.

3.2 Long-Term Prospect on Physics-Aware Optical Flow Estimation

The long-term prospect of this project is present, as deep-learning based flow estimation has still been a vibrant field of research. Recent developments, particularly a newly published study in April 2026, provides valuable insights that align closely with our work, especially for the design of novel loss functions (Sun et al., 2026). Nevertheless, challenges persist in effectively incorporating physical constraints and temporal coherence into flow estimation frameworks.

Hence, if provided sufficient time, resource and attention, the project may still benefit from further effort on the overall performance. In particular, integrating state-of-the-art (SOTA) models that demonstrate superior performance in pairwise flow estimation could significantly enhance the overall system. When combined with our physics and temporal-aware framework, such advancements have the potential to produce more accurate and robust self-supervised learning outcomes.

3.3 From Optical Flow to Scene Flow

In our proposal, we noticed that there has been a trend in expanding optical flow estimation to scene flow prediction. Given that the KITTI dataset does offer disparity

data and ground truth, the development of this technical route is also viable. By incorporating the Neural Operator (NO) modules on a 3D level, it also has the potential to learn the unobservable “orthogonal” components of the flows. This can be done by either applying a 3D version model in the NO family, or simply applying another NO module as the orthogonal component refiner.

Overall, as we have fulfilled our proposed goals, and have achieved certain progress, there still remains spaces for improvement. The case is particularly true for long-term development of the topic.

References

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